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# Analytic Versus Computational Cognitive Models: Agent-Based Modeling as a Tool in Cognitive Sciences

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**Abstract**
Computational cognitive models typically focus on individual behavior in isolation. Models frequently employ closed-form solutions in which a state of the system can be computed if all parameters and functions are known. However, closed-form models are challenged when used to predict behaviors for dynamic, adaptive, and heterogeneous agents. Such systems are complex and typically cannot be predicted or explained by analytical solutions without application of significant simplifications. In addressing this problem, cognitive and social psychological sciences may profitably use agent-based models, which are widely employed to simulate complex systems. We show that these models can be used to explore how cognitive models scale in social networks to calibrate model parameters, to validate model predictions, and to engender model development. Agent-based models allow for controlled experiments of complex systems and can explore how changes in low-level parameters impact the behavior at a whole-system level. They can test predictions of cognitive models and may function as a bridge between individually and socially oriented models.

**Keywords**
agent-based modeling, analytical models, complex systems, computational cognitive models, scientific method

Cognitive methods often elicit data from laboratory experiments, surveys, or similar individual-oriented situations, ensuring controlled settings with little or no experimental noise. Consequently, such computational cognitive theories and models describe interactions in isolation. In addition, because of fear of confounding variables, the context in which data are retrieved and models tested are potentially nonrealistic, as the participant is typically isolated from others, and the experimental conditions are strictly constrained. This is done to isolate the relevant phenomenon and to collect clean data for modeling a specific function (e.g., memory, learning, or decision making). For the purposes of this article, models that use such methods to describe or predict cognitive functions as isolated phenomena are labeled *analytic*.

Analytic models are the bedrock of modeling cognitive functions. They can identify the relevant functions, isolate these experimentally and conceptually, and generate computational models that describe and potentially predict how the functions work, can be modeled, and can be tested. For example, researchers have used Bayesian analytic models to describe, test, and predict the role of argumentation (Hahn & Oaksford, 2006, 2007). Analytic models can be as complicated and expanded as the task requires and can be sufficiently parameterized given data or assumptions.

Yet, independently of how well individual-oriented experiments are designed, most human functions occur in interactions with other people. Crucially, this has two important implications for cognitive models. First, the actions of others may influence how an individual directs his or her attention, learns about a problem, or

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makes decisions. That is, the instantiation of a given cognitive model may depend on interactions with other people. Thus, if researchers wish to test how a given model performs in more realistic systems or contexts, they may need to conceptualize or integrate interactive elements. Second, cognitive scientists may wish to explore how their models perform in social environments (e.g., Sun, 2006). Indeed, cognitive scientists are uniquely placed to provide psychologically realistic models that describe or predict how people perform individually given cognitive components. However, whereas simple interactive systems, such as two-person game-theoretic problems, may be solved analytically, some problems cannot be described or predicted by "multiplying up" each analytic cognitive component in isolation (see Gustafsson & Sternad, 2010). For example, Orr and Plaut (2014) and Orr, Thrush, and Plaut (2016) tested the impact of changes in cognitive computational models on health behavior in complex systems.

Interactions can generate feedback loops that can reverberate through and change the system on a macro scale. Such loops may cause a system to behave unpredictably. Further, such interactions playing out between heterogeneous members of a population (e.g., differences in key cognitive parameters or goals) means it is not possible to scale analytical cognitive models from the single to the plural (Johnson, 2007).

Interactive, nonlinear systems can be complex and thus computationally irreducible. This means that the evolution of their macro state cannot be fully expressed by equations, even if a low-level understanding exists. Therefore, the only way to determine an answer to computationally irreducible problems is to simulate the computation. For the purposes of this article, models that use computational (numerical) solutions to describe or predict cognitive functions as interactive phenomena are labeled *dynamic*.

## The Methodological Challenge

Interactivity poses a challenge for testing computational cognitive models in more realistic contexts: External actions may directly influence cognitive components, and the behavior of cognitive entities is not trivially scaled to the population level. To address this methodological challenge, we present an approach, namely agent-based modeling (Gilbert, 2008), that can implement and test computational cognitive models in dynamic and complex systems. Concretely, we argue that agent-based models (ABMs) provide three advantages for cognitive scientists.

First, ABMs can explore what happens when multiple cognitive entities interact over time and space. They can thus test how cognitive models scale up when they are implemented in social systems. Recently, a number of such simulations and computational models have been used to shed light on important issues in modern societies. For example, using computational cognitive models of belief revision in social networks, we may explore belief diffusion (Duggins, 2017), emergence of echo chambers (Madsen, Bailey, & Pilditch, 2018), belief cascading (Pilditch, 2017), and network pruning (Ngampruetikorn & Stephens, 2016).

Second, ABMs can calibrate and validate cognitive models in complex systems. That is, known aggregate behaviors should emerge from the interacting agents if the cognitive model is appropriate. Moreover, given observed data, ABMs can parameterize model components (calibration) or, given model parameterization, test model predictions against observed data (validation). The use of ABMs to match model predictions against a set of summary statistics has been used in ecology (Hartig, Calabrese, Reineking, & Wiegand Huth, 2011) and economics (Grazzini & Richiardi, 2015). Using ABMs in this way was first suggested by Richiardi, Roberto, Nicole, and Sonnessa (2006) and first implemented by Zhao (2010).

Third, ABMs encourage model development. Whereas we may develop models to deal with isolated cognitive functions (e.g., memory retrieval, belief revision), the act of implementing models in relatively large social systems is useful in explicitly accounting for all aspects of the social system. For example, belief-revision models might require information-search strategies when dealing with social networks (see Madsen et al., 2018), and the nature of these strategies is made explicit in attempts to model them.

Although ABMs have been used in fields such as crowd behavior (Wijermans, Jorna, Jager, Vliet, & Adang, 2013), voter turnout (Fieldhouse, Lessard-Phillips, & Edmonds, 2015), and belief polarization (Pulick, Korth, Grim, & Jung, 2016), they have gained traction only recently in psychology, including organizational psychology (Hughes, Clegg, Robinson, & Crowder, 2012), social ecology (Oishi & Graham, 2010), and social psychology (Smith & Conrey, 2007). Cognitive scientists have begun to use this method (e.g., Duggins, 2017; Madsen et al., 2018; Ngampruetikorn & Stephens, 2016; Pilditch, 2017), although they have not yet done so frequently. We believe that agent-based modeling has the potential to open many new and interesting avenues of research in cognitive science. Though we argue for the advantages of ABMs in this article, we stress that they are merely another tool in the methodological toolbox. If the problem can be solved analytically, there is no reason to employ needlessly complex dynamic models. Thus, we do not argue for the replacement of existing models but rather the addition of a tool for academic problems that are complex in nature.

## What Are ABMs?

ABMs are computer-simulated, multiagent systems that describe the behavior of and interactions between individual agents operating in synthetic environments (Bandini, Manzoni, & Vizzari, 2009; Gilbert, 2008). They have been used in economics (Grazzini & Richiardi, 2015), social sciences (Axtell & Epstein, 1994; Epstein & Axtell, 1996; Schelling, 1971, 2006; see Heath, Hill, & Ciarallo, 2009, for a survey), and ecology (McClane, Semeniuk, McDermid, & Marceau, 2011). They are interactive systems (Bonabeau, 2002) with self-organizing capacities (Niazi & Hussain, 2009).[1] ABMs broadly consist of three elements: *agents, environment*, and *interactions*.

Agents are the individual actors. Agents can be endowed with any cognitive (information-processing) model that is algorithmically expressible and relevant to solving a particular task (e.g., belief revision given new information). ABMs allow for different classes of agents, and parameters can be heterogeneous within a class of agents. This makes ABMs ideal for testing cognitive models in which individual differences are crucial to function.

The environment is the synthetic world in which the agents act or interact. Again, ABMs can include any algorithmically expressible features. Environmental features may enable or disable cognitive tasks (e.g., in models that explore how information is disseminated, technology may increase the reach of people and thus facilitate greater possible contact between agents, whereas geographical distance may limit it). Further, features can be dynamic and may grow, wither, or transform. This allows for dynamic environments in which agents can sample information, learn, and adapt. The environment can also provide impetus to spatially distribute agents if relevant (e.g., communal foraging strategies).

Interactions represent connections between agents. Again, any interaction that is algorithmically expressible can be introduced. Interactions can be direct, indirect, or both. For example, indirectly, agents may prefer or dislike the presence of others, which influences spatial distributions of agents (Schelling, 2006). Directly, agents may transmit information to other agents, causing belief diffusion (Duggins, 2017). In ABMs, interactions may unfold over time (i.e., an agent may contact an agent at time $n$ and a different agent at time $n + 1$). This allows for simulation of the development of behavior and cognitive function as a result of sequential interactions. As discussed above, interactions are likely to make systems complex and therefore in need of simulated solutions (i.e., dynamic models). Together with environmental features, ABMs provide cognitive scientists with methods for exploring and testing cognitive models in complex settings in which any computationally expressible cognitive model can be implemented and tested.

Given the capacity for adaptive strategies and functions in simulated agents, ABMs can generate behaviors that are not hard wired or externally defined. That is, sufficiently adaptive agents can modify their behavior dynamically rather than being forced to "relive" behaviors determined during model construction, such as those informed by data. How agents adapt over time can itself be used to validate cognitive models by assessing whether individual and aggregate behaviors replicate known and observable patterns (e.g., Grimm et al., 2005).

## When Is an ABM Worth Considering as a Methodological Tool?

ABMs are useful when the cognitive function is dependent on the actions of other agents. For example, if darkness falls at 6:32 p.m., we might predict that most cyclists will decide to turn on their light between 6:15 and 6:45 p.m. However, if we wish to model all cyclists navigating the streets of London, their behavior is contingent on decisions of other people in traffic, the road system, and other factors. Statistical extrapolation of isolated individuals may adequately capture individual and aggregate behaviors in the former situation (e.g., a normal distribution of switching on lights). However, individual and aggregate behavior in the latter situation cannot be extrapolated from the components in isolation because of feedback among agents. In Schelling's (2006) term, the latter are *contingent behaviors* that can be modeled by ABMs. Below, we present features that may generally warrant the use of ABMs.

### *Multiple dependent agents*

Many-element systems may be complex when cognitive functions depend on the actions of or information from other agents (e.g., information transmission in social networks). When the system comprises two or more agents who can interact, the system is complex and open to positive and negative feedback loops. Voter behavior is an example of this. Aside from interactions between campaigns and voters, voters also interact with each other (representing bottom-up information structures) during elections periods. Whereas each agent can be expressed analytically (e.g., using Bayesian belief-revision models), the interactions fundamentally influence how each cognitive agent can develop and act. In other words, the structure of the system is an important factor in driving micro and macro behavior in addition to the properties of the individuals. Thus, there are many situations in which, because of the interactions, individual as well as aggregate voting behavior cannot be described or predicted analytically. Rather, in order to understand both aggregate and individual voter behavior, we require simulated solutions provided, for example, by (cognitively informed) ABMs.

### *Heterogeneity within and between groups*

Humans vary within and between groups in terms of individual capabilities, interactions with their environment, and interactions with other people. ABMs readily allow for such heterogeneity. Within a group, parameters can be heterogeneously set in any computationally expressible way (e.g., randomly drawn from a distribution). The parameters can be set theoretically (e.g., defined a priori in conditional probability tables) or empirically (e.g., when parameters of risk aversion are set). Between groups, multiple groups of agents can be described. For example, a financial market may contain regulators and investors (each with their own heterogeneous cognitive models). Although heterogeneity within and between groups represent a challenge for analytic models, both types of heterogeneity are more naturally integrated and their effects more precisely tested using ABMs.

Heterogeneity can be exogenous (e.g., agents start with different parameters) or endogenous (e.g., agents start with the same parameters but diverge over time because of feedback). This allows for differences to be implemented or to emerge as a product of the model (Le Baron, 2006, discusses this for finance ABMs).

### *Passage of time*

Most cognitive functions unfold and change over time. For some actions, there is path dependence, in which agents learn about the environment and adopt new strategies to cope with a task. Time also allows for the emergence of group dynamics such as collaboration or defection (suggesting, for example, the growth of cultural practices). ABMs naturally simulate the passage of time, as the various states in the model are sequentially updated as the simulation progresses.

### *Spatial distribution*

Spatial placement is crucial to some cognitive tasks, such as navigating traffic or foraging for spatially distributed resources. Given that the environment is defined explicitly in the model and that agents can operate within the space, ABMs naturally allow for spatial effects and interactions. This may result in identifiable spatial dynamics or adaptive strategies that relate specifically to spatial distribution of agents (e.g., people fishing along the edges of marine protected areas to catch biomass that emerges from those areas; Holland, 2008). ABMs are well suited to tasks involving space and time.

ABMs are useful when the computational cognitive model is concerned with multiple interacting agents, heterogeneity between or within agent groups, the passage of time, or spatial distribution. Because these features are prevalent in most, if not all, cognitive functions in real life, we argue that ABMs should be considered as a viable tool for cognitive scientists.

## What Are the Uses of ABMs in Cognitive Sciences?

Having considered when ABMs may be appropriate, we now present the uses of ABMs for cognitive scientists. Aside from exploring cognitive models in complex scenarios (as discussed previously), ABMs can calibrate and validate or test cognitive models, run controlled and repeatable simulated experiments of key variables, test longitudinal models, and run experiments that would be unethical if conducted on real people.

### *Calibration and validation*

An important goal for cognitive science is to understand how humans acquire knowledge, make decisions, and solve problems in the real world. This involves an exploration of how cognitive models scale on societal levels. As part of this exploration, cognitive scientists must gain knowledge about the parameters of their models so they are able to better choose between the appropriateness of alternative cognitive models. If observational data are available for individual or aggregate levels, ABMs can be used to calibrate model parameters or to validate and test model predictions. For calibration, ABM results can be compared with known data to set parameters to best replicate and reproduce the data. Calibration tests whether the computational model can be retroactively parameterized to account for observed data. The use of ABMs to calibrate cognitive models has been demonstrated in previous studies (e.g., Grazzini & Richiardi, 2015; Zhao, 2010; see Lee et al., 2015, for a discussion of model outputs).

Whereas calibration is focused on parameter setting, validation tests whether the model accounts for observed data given the chosen model parameters. For validation, parameters may be set theoretically, empirically, or via calibration. Once the model is parameterized, simulations are conducted that can be compared with observed data. If the parameterized model cannot replicate observed individual or aggregate behaviors, it indicates potential problems with the model, environment specifications, social links, the data, or any combination thereof. Calibration provides relevant parameterization; validation tests model predictions.

### *Longitudinal and ethical questions*

Some cognitive models describe long-term functions, such as evolutionary cognitive psychology. Although evolutionary theories are difficult to test empirically, evolutionary principles can be tested and explored longitudinally in an ABM. Further, ABMs can explore interventions and hypotheses that would be unethical in real-life experiments, as “participants” in ABMs are simulated rather than real.

Although ABMs potentially offer novel modeling options for cognitive scientists, analytical models may be preferable when they are applicable. That is, if the conditions for successful application of closed-form solutions exist, such solutions reveal the full possible scope of model response, which provides deeper insights than unique numerical solutions. However, for cognitive functions at work within complex social systems, it is unlikely that analytical models with closed-form solutions would be possible. In essence, it is precisely the lack of the necessary simplification in the real world that would be required to achieve such solutions that means numerical solutions are necessary. We believe that the recognition of this alternative is crucial to the development and testing of adequate cognitive models that account for anything other than highly simplified operational conditions.

## Concluding Remarks

Cognitive functions unfolding in complex systems cannot be solved analytically—addressing them requires a different set of tools. We have argued that agent-based modeling provides a valuable tool for exploring, testing, and developing cognitive models in such systems. Simulations can be used to explore theoretical questions (e.g., How does a cognitive model scale up in a social network?), to calibrate model parameters, to validate model predictions, and to develop model components. Additionally, they allow for adaptive agents changing over time and heterogeneous populations, and they may inspire empirical testing of specific hypotheses generated through simulations (cf. Gilbert & Troitzsch, 2005; Holm, Lemm, Thees, & Hilty, 2016; Janssen & Ostrom, 2006; Wunder, Suri, & Watts, 2013).

Finally, ABMs potentially provide a methodological framework to span the gap between individual-level cognitive models and population-level sociological questions. There is no grand unifying theory on how to go from understanding individuals to understanding populations. ABMs provide a framework to explore and possibly test cognitive models in large-scale social systems. In addition, they may provide a computational method for management of complex and dynamic human-environment systems (Bailey et al., 2019) to avoid problems such as the tragedy of the commons (Ostrom, 2012).

### Recommended Reading

Heath, B., Hill, R., & Ciarallo, F. (2009). (See References). An overview of methodologies used for developing agent-based models, an analysis of the proliferation of the method in various publications, and a discussion of validation techniques for researchers considering using agent-based models.

Johnson, N. (2007). (See References). A discussion of the conceptual foundation for complex systems that provides a good introduction to readers who are unfamiliar with complexity as a theory and a historical discipline.

Miller, J. H., & Page, S. E. (2007). (See References). An in-depth description of how agents can self-organize and adapt in dynamic, complex systems.

Müller, B., Bohn, F., Dreßler, G., Groeneveld, J., Klassert, C., Martin, R., . . . Schwarz, N. (2013). (See References). A description of a methodological tool to describe a model in clear and contextual principles.

Wilensky, U., & Rand, W. (2015). (See References). An introduction for people who want to learn how to build agent-based models in NetLogo (the programming language frequently used for these types of models).

### Action Editor

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The author(s) declared that there were no conflicts of interest with respect to the authorship or the publication of this article.

### Note

1. ABMs can be developed in multiple languages; for example, the multiagent simulator of neighborhoods (MASON) toolkit (Luke, Cioffi-Revilla, Panait, Sullivan, & Gabriel, 2005) and NetLogo (Wilensky & Rand, 2015). For model protocols, see Polhill, Parker, Brown, and Grimm (2008), Grimm et al. (2010), and Müller et al. (2013). See Miller and Page (2007, Appendix B) for a list of methodological questions to consider.

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